

CL 9_ MATHS_ PAPER 2 _ HALF YEARLY

Date: 06/09/2021

Duration: 45 minutes

Marks: 30

The respondent's email (**null**) was recorded on submission of this form.

*** Required**

1. Email *

2. NAME: *

CLASS 9:

3. SECTION: *

Mark only one oval.

☐ A

☐ B

☐ C

☐ D

☐ E

☐ F

☐ G

☐ H

☐ I

☐ J

4. Question 1 *

2 points

1) The number obtained on rationalizing the denominator of $\frac{1}{\sqrt{5}-2}$

Mark only one oval.

a) $\frac{\sqrt{5}+2}{21}$

☐ Option 1

b) $\sqrt{5} + 2$

☐ Option 2

c) $\sqrt{5} - 2$

☐ Option 3

d) $\frac{\sqrt{5}-2}{21}$

☐ Option 4

5. Question 2. Find the Compound Interest for second year on ₹ 25000 at 12% per annum compounded annually. * 2 points

Mark only one oval.

- ☐ ₹3000
- ☐ ₹3600
- ☐ ₹3360
- ☐ ₹3060

6. Question 3 * 2 points

3) If $a + b + 2c = 0$, then the value of $a^3 + b^3 + 8c^3$ is

Mark only one oval.

- ☐ abc
- ☐ 2abc
- ☐ 4abc
- ☐ 6abc

7. Question 4 *

2 points

4) After factorization $3 - 12(a - b)^2$ becomes

Mark only one oval.

☐ $3(1 + 2a - 2b)(1 - 2a + 2b)$

☐ $(3 + 6a - 6b)(3 - 6a - 6b)$

☐ $3(1 - 2a - 2b)(1 - 2a + 2b)$

☐ $(3 + 6a - 6b)(3 - 6a + 6b)$

8. Question 5. The value of x for the pairs of equations, $3x + 2y = 4$ and $8x + 5y = 9$ is *

2 points

Mark only one oval.

☐ 2

☐ - 2

☐ 5

☐ -5

9. Question 6 *

2 points

6) The value of $(a^{-1} + b^{-1})(a + b)^{-1}$ is

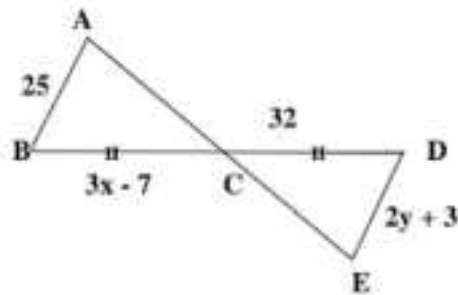
Mark only one oval.

- ☐ ab
- ☐ $1 \div (ab)$
- ☐ $1 \div a$
- ☐ $1 \div b$

10. Question 7 *

2 points

7) In the following figure if $\angle BAC = \angle CED$, calculate the value of x and y



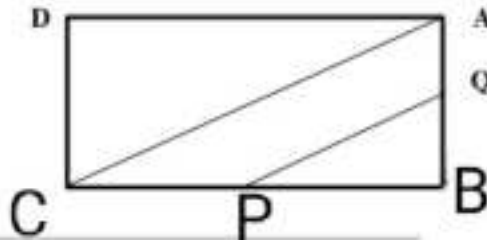
Mark only one oval.

- ☐ $x = 13, y = 11$
- ☐ $x = 11, y = 13$
- ☐ $x = -13, y = 11$
- ☐ $x = -11, y = 13$

11. Question 8 *

2 points

- 8) In the adjoining figure, ABCD is a rectangle in which $AB = 6\text{cm}$ and $AD = 8\text{cm}$. If P and Q are mid points of the sides BC and AB respectively, then the length of PQ is



Mark only one oval.

- ☐ 7cm
- ☐ 5cm
- ☐ 4cm
- ☐ 3cm

12. Question 9 *

2 points

- 9) If $\cos A = \frac{4}{5}$, then the value of $\tan A$ is

Mark only one oval.

- ☐ 3/5
- ☐ 3/4
- ☐ 4/3
- ☐ 5/3

13. Question 10. The distance between C (-1, 4) and D (-2 , -1) is *

2 points

Mark only one oval.

- ☐ $\sqrt{34}$ units
- ☐ $\sqrt{18}$ units
- ☐ $\sqrt{26}$ units
- ☐ $\sqrt{10}$ units

14. Question 11. How many times the wheel of a car can rotate in a journey of 88km if it is known that the diameter of the wheel is 56cm. *

2 points

Mark only one oval.

- ☐ 50
- ☐ 500
- ☐ 5000
- ☐ 50000

15. Question 12 *

2 points

12) The value of $2\tan 30^\circ \div (1 + \tan^2 30^\circ)$ is equal to,

Mark only one oval.

- ☐ $\cos 60^\circ$
- ☐ $\sin 60^\circ$
- ☐ $\tan 60^\circ$
- ☐ $\operatorname{cosec} 60^\circ$

16. Question 13. *

2 points

13) If $x - \frac{1}{x} = 5$, then the value of $x^2 + (\frac{1}{x})^2$ is

Mark only one oval.

☐ 25

☐ 23

☐ 27

☐ 29

17. Question 14 *

2 points

14) Factorisation of $x^2 - 4x - 12$ is

Mark only one oval.

☐ $(x + 6)(x - 2)$

☐ $(x - 6)(x + 2)$

☐ $(x - 6)(x - 2)$

☐ $(x + 6)(x + 2)$

18. Question 15 *

2 points

15) If $a = 3$ and $b = -2$, then the value of $a^b + b^a$ is

Mark only one oval.

☐ - 71/9

☐ 17

☐ 1

☐ 12

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